4R Robot Arm

Kinematics and Control of a 4-Joint Arm

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Abstract

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Notation

Symbol	Meaning
$\boldsymbol{q}$	
${}^{i-1}\underline{T}_i$	

Integration variables are bound to their integral, as in programming languages: in an expression such as

$$s(t) = \int_0^t dt \, \|\dot{\boldsymbol{\gamma}}(t)\|,$$

the t inside the integral is the integration variable and is independent of the upper limit t.

1. Introduction

1.1. Motivation

1.2. Goals and scope

1.3. Structure of this document

2. Prerequisites

The book Mathematik [1] can be used to study all the math required for this project.

2.1. Fundamentale physikalische Größen

1. Impuls.

$$\boldsymbol{p} = m\boldsymbol{v}$$

2. Kraft.

$$\boldsymbol{F} = \frac{d\boldsymbol{p}}{dt}, \qquad \boldsymbol{F} = m\boldsymbol{a}$$

3. Drehimpuls.

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} = m\,\boldsymbol{r} \times \boldsymbol{v} = m\,(\boldsymbol{r} \cdot \boldsymbol{r})\,\boldsymbol{\omega}, \qquad \text{unter der Annahme } \langle \boldsymbol{r}, \boldsymbol{\omega} \rangle = 0.$$

Der Drehimpuls hängt von der Wahl des Bezugspunktes ab. Meint man allerdings den Drehimpuls eines Körpers, ist dieser Bezugspunkt eindeutig als der Schwerpunkt festgelegt.

4. Drehmoment.

$$\boldsymbol{M} = \frac{d\boldsymbol{L}}{dt}, \qquad \boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F}$$

Das Drehmoment verhält sich zum Drehimpuls wie die Kraft zum Impuls.

5. Trägheitsmoment. Das Trägheitsmoment ist definiert als eine Größe, die natürlich in den Formeln für kinetische Energie, Drehimpuls und Drehmoment auftritt. Hier werden die Größen für eine Scheibe beschrieben, die in der x-y-Ebene liegt und sich um die z-Achse dreht. r gibt dabei den Abstand zur z-Achse an und ρ die Flächendichte. ω und α sind die Winkelgeschwindigkeit und die Winkelbeschleunigung der Scheibe, $v(r) = \omega r$ und $a(r) = \alpha r$ die Bahngeschwindigkeit und die Bahnbeschleunigung im Abstand r, und

$$I = \iint \rho(r,\varphi)\, r^2\, r\, dr\, d\varphi.$$

Damit gilt

$$\begin{aligned}
L &= \iint \rho(r,\varphi)\, v(r)\, r\; r\, dr\, d\varphi = \iint \rho(r,\varphi)\, \omega r^2\; r\, dr\, d\varphi = \omega I, \\
E_{\text{kin}} &= \frac{1}{2} \iint \rho(r,\varphi)\, v(r)^2\; r\, dr\, d\varphi = \frac{1}{2} \iint \rho(r,\varphi)\, \omega^2 r^2\; r\, dr\, d\varphi = \frac{1}{2}\omega^2 I, \\
M &= \iint \rho(r,\varphi)\, a(r)\, r\; r\, dr\, d\varphi = \iint \rho(r,\varphi)\, \alpha r^2\; r\, dr\, d\varphi = \alpha I.
\end{aligned}$$

Diese Beschreibungen sind allgemein und hilfreich für nicht deformierbare Scheiben. Das Trägheitsmoment nimmt eine ähnliche Rolle ein wie die Masse.

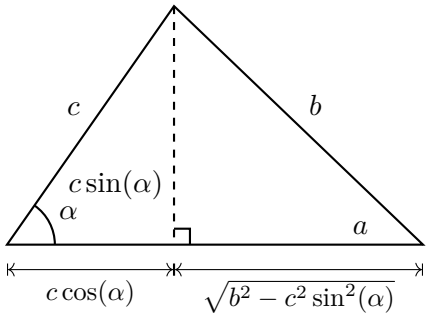


Figure 2.1.: Triangle with acute angle α. The altitude onto a divides a into two segments.

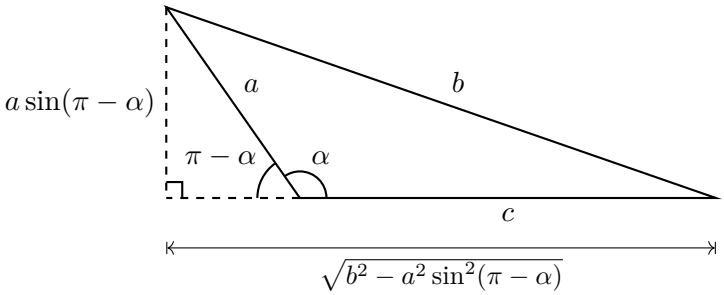


Figure 2.2.: Triangle with obtuse angle α. The altitude falls onto the extension of c.

2.2. Laws for sine and cosine

Lemma 2.1 (Laws for sine and cosine)**.** *Let a and c be the lengths of the edges adjacent to the angle α of a triangle, and let b be the length of the edge opposite to α. Then*

$$\cos(\alpha) = \frac{a^2 + c^2 - b^2}{2ac}, \tag{2.1}$$

$$\sin(\alpha) = \frac{1}{2ac}\sqrt{(a+c+b)(a+c-b)(a-c+b)(a-c-b)}. \tag{2.2}$$

Proof. Cosine. If α is acute (fig. 2.1), the altitude onto the edge a has length $c\sin(\alpha)$ and divides a into two segments, so that

$$\begin{aligned} a &= c\cos(\alpha) + \sqrt{b^2 - c^2\sin^2(\alpha)} \\ \Rightarrow \quad a^2 - 2ac\cos(\alpha) + c^2\cos^2(\alpha) &= b^2 - c^2\sin^2(\alpha) \\ \Rightarrow \quad a^2 - 2ac\cos(\alpha) &= b^2 - c^2 \\ \Rightarrow \quad \cos(\alpha) &= \frac{a^2 + c^2 - b^2}{2ac}. \end{aligned}$$

If α is obtuse (fig. 2.2), the altitude falls outside the triangle, onto the extension of the edge c, and has length $a\sin(\pi - \alpha)$. Then

$$
\begin{aligned}
c &= \sqrt{b^2 - a^2\sin^2(\pi-\alpha)} - a\cos(\pi-\alpha) \\
&= \sqrt{b^2 - a^2\sin^2(\alpha)} + a\cos(\alpha) \\
\Rightarrow \quad c^2 - 2ac\cos(\alpha) + a^2\cos^2(\alpha) &= b^2 - a^2\sin^2(\alpha) \\
\Rightarrow \quad c^2 - 2ac\cos(\alpha) &= b^2 - a^2 \\
\Rightarrow \quad \cos(\alpha) &= \frac{a^2 + c^2 - b^2}{2ac}.
\end{aligned}
$$

In general this holds for any of the three angles of the triangle: the lengths that can be interchanged without changing the result are always those of the two adjacent edges, and the remaining length is that of the opposite edge.

Sine. Using (2.1),

$$
\begin{aligned}
\sin(\alpha) &= \sqrt{1 - \cos^2(\alpha)} = \sqrt{(1+\cos(\alpha))(1-\cos(\alpha))} \\
&= \sqrt{\left(1 + \frac{a^2+c^2-b^2}{2ac}\right)\left(1 - \frac{a^2+c^2-b^2}{2ac}\right)} \\
&= \sqrt{\left(\frac{(a+c)^2-b^2}{2ac}\right)\left(\frac{(a-c)^2-b^2}{2ac}\right)} \\
&= \frac{1}{2ac}\sqrt{(a+c+b)(a+c-b)(a-c+b)(a-c-b)}.
\end{aligned}
$$

□

2.3. Curvature

This section on curvature closely follows Wikipedia [2].

Definition.

$$
\kappa = \left\| \frac{d\boldsymbol{T}}{ds} \right\|, \tag{2.3}
$$

where s denotes the arc length and $\boldsymbol{T}$ the normalized tangent vector,

$$
\|\boldsymbol{T}\| = 1 \quad \text{for all } s \in [0, L],
$$

where L describes the length of the curve or the length at which the curve repeats ($2\pi r$ for a circle with radius r).

Since every tangent vector is normalized, we can express it in the form

$$
\boldsymbol{T}(s) = \begin{pmatrix} \cos(\varphi(s)) \\ \sin(\varphi(s)) \end{pmatrix},
$$

where $\varphi(s)$ is the angle of the tangent vector. Therefore we find

$$
\kappa = \left\| \frac{d\boldsymbol{T}(s)}{ds} \right\| = \left\| \begin{pmatrix} -\sin(\varphi(s)) \\ \cos(\varphi(s)) \end{pmatrix} \frac{d\varphi(s)}{ds} \right\| = \left| \frac{d\varphi}{ds} \right|.
$$

If a curve $\boldsymbol{\gamma}(t)$ is not parametrized in terms of the arc length, we can find its arc-length parametrization $\bar{\boldsymbol{\gamma}}(s)$ via

$$
s(t) = \int_0^t dt\, \|\dot{\boldsymbol{\gamma}}(t)\|
$$

and therefore

$$\frac{ds}{dt} = \|\dot{\boldsymbol{\gamma}}(t)\| \quad \text{and} \quad \frac{dt}{ds} = \frac{1}{\|\dot{\boldsymbol{\gamma}}(t)\|}.$$

With those we find

$$\frac{d\bar{\boldsymbol{\gamma}}(s)}{ds} = \frac{d\boldsymbol{\gamma}(t(s))}{ds} = \frac{\partial \boldsymbol{\gamma}}{\partial t}\frac{dt}{ds} = \dot{\boldsymbol{\gamma}}(t)\frac{1}{\|\dot{\boldsymbol{\gamma}}(t)\|}.$$

Since $d\bar{\boldsymbol{\gamma}}/ds$ is the tangent vector, we can also immediately verify the normalization of the tangent vector.

One can also use the chain rule again to find the curvature in the non-arc-length parametrization:

$$\kappa = \left\| \frac{d\boldsymbol{T}}{ds} \right\|$$

with

$$\begin{aligned}
\frac{d\boldsymbol{T}}{ds} &= \frac{d}{ds}\left(\frac{\dot{\boldsymbol{\gamma}}(t(s))}{\|\dot{\boldsymbol{\gamma}}(t(s))\|}\right) \\
&= \frac{\ddot{\boldsymbol{\gamma}}(t(s))}{\|\dot{\boldsymbol{\gamma}}(t)\|}\cdot\frac{1}{\|\dot{\boldsymbol{\gamma}}(t)\|} - \frac{1}{2}\frac{\dot{\boldsymbol{\gamma}}(t)}{\left(\dot{\boldsymbol{\gamma}}(t)\cdot\dot{\boldsymbol{\gamma}}(t)\right)^{3/2}}\cdot 2\,\dot{\boldsymbol{\gamma}}(t)\cdot\ddot{\boldsymbol{\gamma}}(t)\cdot\frac{1}{\|\dot{\boldsymbol{\gamma}}(t)\|} \\
&= \frac{\ddot{\boldsymbol{\gamma}}(t)}{\|\dot{\boldsymbol{\gamma}}(t)\|^2} - \dot{\boldsymbol{\gamma}}(t)\big(\dot{\boldsymbol{\gamma}}(t)\cdot\ddot{\boldsymbol{\gamma}}(t)\big)\frac{1}{\|\dot{\boldsymbol{\gamma}}(t)\|^4} \\
&= \frac{\ddot{\boldsymbol{\gamma}}(t)}{\|\dot{\boldsymbol{\gamma}}(t)\|^2} - \boldsymbol{T}\big(\boldsymbol{T}\cdot\ddot{\boldsymbol{\gamma}}(t)\big)\frac{1}{\|\dot{\boldsymbol{\gamma}}(t)\|^2}.
\end{aligned}$$

Therefore

$$\begin{aligned}
\kappa = \left\|\frac{d\boldsymbol{T}}{ds}\right\| &= \sqrt{\frac{\|\ddot{\boldsymbol{\gamma}}(t)\|^2}{\|\dot{\boldsymbol{\gamma}}(t)\|^4} + \frac{\left(\boldsymbol{T}\cdot\ddot{\boldsymbol{\gamma}}(t)\right)^2}{\|\dot{\boldsymbol{\gamma}}(t)\|^4} - 2\,\frac{\left(\boldsymbol{T}\cdot\ddot{\boldsymbol{\gamma}}(t)\right)^2}{\|\dot{\boldsymbol{\gamma}}(t)\|^4}} \\
&= \sqrt{\frac{\|\ddot{\boldsymbol{\gamma}}(t)\|^2\|\dot{\boldsymbol{\gamma}}(t)\|^2}{\|\dot{\boldsymbol{\gamma}}(t)\|^6} - \frac{\left(\dot{\boldsymbol{\gamma}}(t)\cdot\ddot{\boldsymbol{\gamma}}(t)\right)^2}{\|\dot{\boldsymbol{\gamma}}(t)\|^6}} \\
&= \frac{\sqrt{\|\ddot{\boldsymbol{\gamma}}(t)\|^2\|\dot{\boldsymbol{\gamma}}(t)\|^2 - \left(\dot{\boldsymbol{\gamma}}(t)\cdot\ddot{\boldsymbol{\gamma}}(t)\right)^2}}{\|\dot{\boldsymbol{\gamma}}(t)\|^3}.
\end{aligned} \tag{2.4}$$

2.3.1. Special curvatures

1. Circle. A circle is typically parametrized by the angle θ:

$$\boldsymbol{\gamma}(\theta) = \begin{pmatrix} r\cos(\theta) \\ r\sin(\theta) \end{pmatrix}.$$

Its arc length is

$$s(\theta) = \int_0^\theta d\theta \left\| \begin{pmatrix} -r\sin(\theta) \\ r\cos(\theta) \end{pmatrix} \right\| = r\theta, \qquad \text{so} \qquad \theta(s) = \frac{s}{r},$$

and we find the curve parametrized with the arc length s:

$$\boldsymbol{\gamma}(\theta(s)) = \begin{pmatrix} r\cos\left(\frac{s}{r}\right) \\ r\sin\left(\frac{s}{r}\right) \end{pmatrix}, \qquad \boldsymbol{T} = \frac{d\boldsymbol{\gamma}}{ds} = \begin{pmatrix} -\sin\left(\frac{s}{r}\right) \\ \cos\left(\frac{s}{r}\right) \end{pmatrix}, \qquad \frac{d\boldsymbol{T}}{ds} = \begin{pmatrix} -\frac{1}{r}\cos\left(\frac{s}{r}\right) \\ -\frac{1}{r}\sin\left(\frac{s}{r}\right) \end{pmatrix},$$

and therefore the curvature

$$\kappa(s) = \left\| \frac{d\boldsymbol{T}(s)}{ds} \right\| = \frac{1}{r}.$$

2. Graph of a function. For the graph of a function f, where dots denote derivatives with respect to x,

$$\boldsymbol{\gamma}(x) = \begin{pmatrix} x \\ f(x) \end{pmatrix}, \qquad \dot{\boldsymbol{\gamma}}(x) = \begin{pmatrix} 1 \\ \dot{f}(x) \end{pmatrix}, \qquad \ddot{\boldsymbol{\gamma}}(x) = \begin{pmatrix} 0 \\ \ddot{f}(x) \end{pmatrix}.$$

Now we can use the general parametrization formula (2.4):

$$\kappa = \frac{\sqrt{\left(\ddot{f}(x)\right)^2 \left(1 + (\dot{f}(x))^2\right) - \left(\dot{f}(x)\, \ddot{f}(x)\right)^2}}{\left(1 + (\dot{f}(x))^2\right)^{3/2}},$$

$$\kappa = \frac{|\ddot{f}(x)|}{\left(1 + (\dot{f}(x))^2\right)^{3/2}}.$$

3. Generalization of a circle. With a radius $r(\theta)$ that depends on the angle, where dots denote derivatives with respect to θ and the argument θ of the trigonometric functions is omitted on the right,

$$\boldsymbol{\gamma}(\theta) = \begin{pmatrix} r(\theta)\cos(\theta) \\ r(\theta)\sin(\theta) \end{pmatrix},$$

$$\dot{\boldsymbol{\gamma}}(\theta) = \begin{pmatrix} \dot{r}\cos - r\sin \\ \dot{r}\sin + r\cos \end{pmatrix},$$

$$\ddot{\boldsymbol{\gamma}}(\theta) = \begin{pmatrix} \ddot{r}\cos - 2\dot{r}\sin - r\cos \\ \ddot{r}\sin + 2\dot{r}\cos - r\sin \end{pmatrix},$$

with

$$\begin{aligned} \|\dot{\boldsymbol{\gamma}}(\theta)\|^2 &= \dot{r}^2 + r^2, \\ \|\ddot{\boldsymbol{\gamma}}(\theta)\|^2 &= \ddot{r}^2 + r^2 + 4\dot{r}^2 - 2\ddot{r}r, \\ (\dot{\boldsymbol{\gamma}} \cdot \ddot{\boldsymbol{\gamma}}) &= \ddot{r}\dot{r} + r\dot{r}. \end{aligned}$$

We find

$$\begin{aligned} \|\dot{\boldsymbol{\gamma}}(\theta)\|^2 \|\ddot{\boldsymbol{\gamma}}(\theta)\|^2 &= \ddot{r}^2\dot{r}^2 + 4\dot{r}^4 + r^2\dot{r}^2 - 2r\ddot{r}\dot{r}^2 + r^2\ddot{r}^2 + 4\dot{r}^2r^2 + r^4 - 2r^3\ddot{r}, \\ \left(\dot{\boldsymbol{\gamma}} \cdot \ddot{\boldsymbol{\gamma}}\right)^2 &= \ddot{r}^2\dot{r}^2 + 2\ddot{r}\dot{r}^2 r + r^2\dot{r}^2, \\ \|\dot{\boldsymbol{\gamma}}(\theta)\|^2 \|\ddot{\boldsymbol{\gamma}}(\theta)\|^2 - \left(\dot{\boldsymbol{\gamma}}(\theta) \cdot \ddot{\boldsymbol{\gamma}}(\theta)\right)^2 &= 4\dot{r}^4 - 4\ddot{r}\dot{r}^2 r + r^2\ddot{r}^2 + 4\dot{r}^2r^2 + r^4 - 2r^3\ddot{r} \\ &= \left(r^2 + 2\dot{r}^2 - r\ddot{r}\right)^2, \end{aligned}$$

so we find

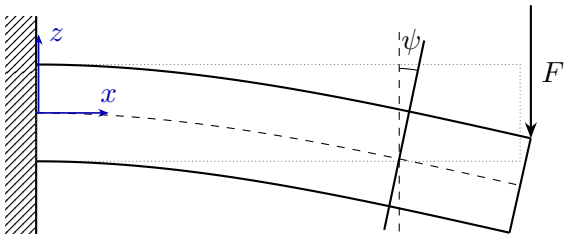
$$\kappa = \frac{\sqrt{\|\dot{\boldsymbol{\gamma}}(\theta)\|^2 \|\ddot{\boldsymbol{\gamma}}(\theta)\|^2 - \left(\dot{\boldsymbol{\gamma}}(\theta) \cdot \ddot{\boldsymbol{\gamma}}(\theta)\right)^2}}{\|\dot{\boldsymbol{\gamma}}(\theta)\|^3} = \frac{r^2 + 2\dot{r}^2 - r\ddot{r}}{\left(r^2 + \dot{r}^2\right)^{3/2}}.$$

2.4. Lineare Balkentheorie

Die alleinige Quelle für diesen Abschnitt ist das Buch TM 2 by Gross [3]. Wer mehr als nur eine kurze Aufffrischung braucht sollte das Kapitel zur Balkenbiegung zu Rate ziehen. Dort sind die Zusammenhänge auch sicherlich besser dargestellt und mehr Aufwand ist in die einwandfreie Visualisierung der Inhalte geflossen.

Motivation. Wir wollen sicherstellen, dass sich der Arm nur minimal unter seiner eigenen Last und dem zu hebenden Gewicht verformt. Da wir somit sehr kleine Verformungen voraussetzen, ist die lineare Balkentheorie ausreichend. Eine mögliche Schlussfolgerung aus konkreten Berechnungen kann allerdings sein, dass die maximal zulässige Last für eine bestimmte kleine Verformung im realistischen Betrieb überschritten wird. Möchte man dennoch weder die Konstruktion des Arms noch die zulässige Last anpassen, muss man die Verformungen in höheren Ordnungen entwickeln und dann die Position des Arms entsprechend der Verformung korrigieren. Letzteres ist allerdings sicherlich ein schwieriges Unterfangen und man wäre wohl besser beraten, wo möglich die Konstruktion anzupassen.

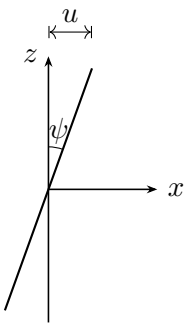
Szenario.



Die Deformation des Balkens ist hier zu Gunsten der Anschaulichkeit übertrieben dargestellt. *Annahme:* Die mittlere gestrichelte Linie beschreibt die Linie, welche ihre Länge als Konsequenz der Verformung nicht verändert. u beschreibt die Verschiebung parallel zu dieser Linie, w beschreibt die Verschiebung vertikal zu $u(s,t)$, aber immer noch in der skizzierten Ebene liegend.

Da wir sehr kleine Verformungen annehmen, sind die s- und t-Achse beinahe parallel zu jeweils der x- und z-Achse.

Betrachten wir nun den skizzierten Querschnitt an Stelle s:



Dann gilt offensichtlich

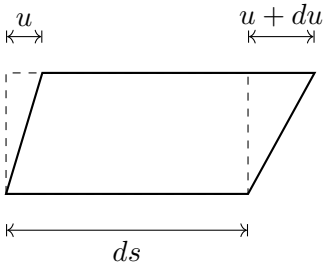
$$\psi(s) = \arctan\left(\frac{u}{z}\right).$$

Da wir annehmen, dass $\psi(s)$ sehr klein ist, muss auch u/z sehr klein sein:

$$\psi(s) \approx \frac{u}{z},$$

$$\boxed{u(s,z) = \psi(s)\, z} \tag{2.5}$$

Betrachten wir nun ein infinitesimales Element und stellen uns vor, dass dieses Element aus horizontal gekoppelten Federn besteht:



Dann beschreibt $du(z)$ genau die Streckung bzw. Stauchung dieser Federn und $\frac{du}{ds}(s, z)$ beschreibt die Stauchung durch die Länge ds der entspannten Feder.

Es lässt sich experimentell das Elastizitätsmodul E bestimmen:

$$\boxed{\sigma = E\varepsilon = E\frac{du}{ds}} \tag{2.6}$$

Mit der Approximation für die Verschiebung u (2.5) finden wir

$$\sigma = E\psi'(s)\, z.$$

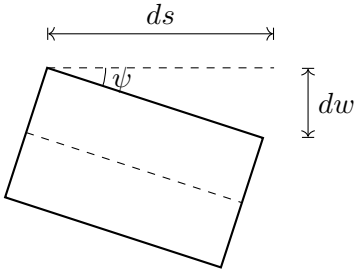
Mit Hilfe der Spannung σ lässt sich der resultierende Moment berechnen, wobei der Bezugspunkt der Ursprung des lokalen Koordinatensystems ist und in der skizzierten Ebene gegen den Uhrzeigersinn rotiert wird:

$$M(s) = \int \sigma(s, z)\, z\, dA = E\psi'(s) \int z^2\, dA,$$

$$M(s) = EI\psi'(s).$$

Dabei ist A die Querschnittsfläche und $I = \int z^2\, dA$ das Flächenträgheitsmoment des Querschnitts.

Wir können $\psi(s)$ durch die vertikale Verschiebung ausdrücken:



$$\psi(s) = \arctan\left(\frac{dw}{ds}\right) \approx \frac{dw}{ds}$$

Hier ist nicht zu vergessen, dass wir eine Verformung beschreiben, die so klein ist, dass die Verformung w, die eigentlich senkrecht auf der Nullspannungslinie steht, auch beinahe vertikal ist.

Mit dieser Approximation finden wir eine Differentialgleichung für die vertikale Verformung w:

$$\boxed{\frac{d^2w}{ds^2}(s) = \frac{M(s)}{EI}} \tag{2.7}$$

Für eine vertikale Kraft F am Ende des Stabes bei Position L gilt für $M(s)$:

$$M(s) = F(L - s)$$

und daher

$$w(s) = \frac{s^2\left(FL - \frac{Fs}{3}\right)}{2EI}$$

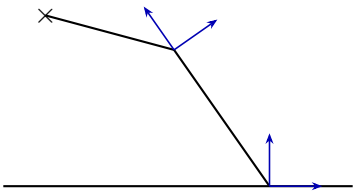
und am Ende des Balkens

$$w(L) = \frac{FL^3}{3EI}.$$

3. Forward Kinematics

3.1. 2R

The arm is treated in the plane of its two links, with y the horizontal and z the vertical coordinate. R_1 and R_2 are the lengths of the first and second link, φ_1 and φ_2 are the joint angles, and positions are written as $(1, y, z)^\mathsf{T}$, so that translations can be expressed as matrix products as well. The primes on $\boldsymbol{r}_2$ indicate the basis in which it is expressed.

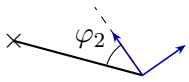


First determine the position $\boldsymbol{r}_2$ in the basis attached to the second link:



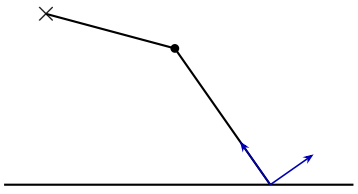
$$\boldsymbol{r}_2''' = \begin{pmatrix} 1 \\ 0 \\ R_2 \end{pmatrix}$$

Change basis to the basis where the z axis aligns with the next link.



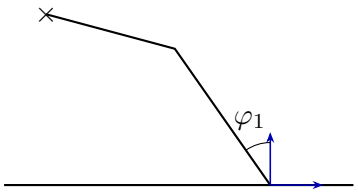
$$\boldsymbol{r}_2'' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi_2 & -\sin\varphi_2 \\ 0 & \sin\varphi_2 & \cos\varphi_2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ R_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -R_2 \sin\varphi_2 \\ R_2 \cos\varphi_2 \end{pmatrix}$$

Change basis to the basis that is still aligned with the link but sits at the bottom joint. This makes the base change a mere translation.

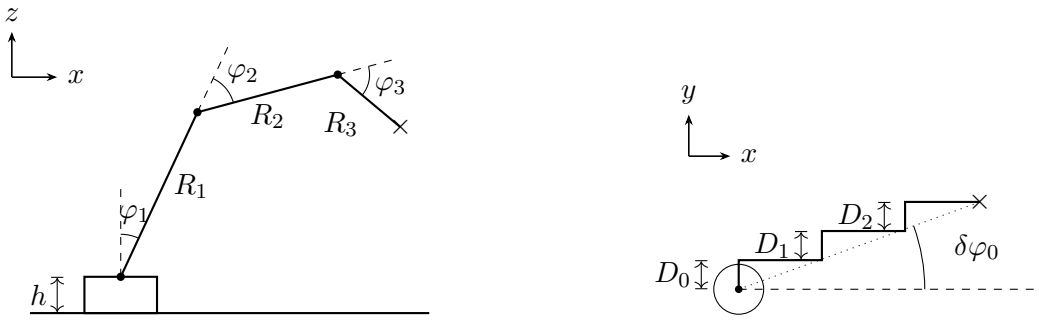


$$\boldsymbol{r}_2' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ R_1 & 0 & 1 \end{pmatrix} \boldsymbol{r}_2'' = \begin{pmatrix} 1 \\ -R_2 \sin\varphi_2 \\ R_2 \cos\varphi_2 + R_1 \end{pmatrix}$$

Change basis to standard y-z basis with origin at base joint.



$$\boldsymbol{r}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi_1 & -\sin\varphi_1 \\ 0 & \sin\varphi_1 & \cos\varphi_1 \end{pmatrix} \boldsymbol{r}_2'$$



(a) Side view. (b) Top view, $\varphi_0 = 0$.

Figure 3.1.: Assumed general structure of the 4R arm.

$$\begin{aligned}\boldsymbol{r}_2 &= \begin{pmatrix} 1 \\ -R_2\cos\varphi_1\sin\varphi_2 - R_2\cos\varphi_2\sin\varphi_1 - R_1\sin\varphi_1 \\ -R_2\sin\varphi_1\sin\varphi_2 + R_2\cos\varphi_1\cos\varphi_2 + R_1\cos\varphi_1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ -R_1\sin\varphi_1 - R_2\sin(\varphi_1+\varphi_2) \\ R_1\cos\varphi_1 + R_2\cos(\varphi_1+\varphi_2) \end{pmatrix}\end{aligned}$$

The entire transformation:

$$\boldsymbol{r}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi_1 & -\sin\varphi_1 \\ 0 & \sin\varphi_1 & \cos\varphi_1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ R_1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi_2 & -\sin\varphi_2 \\ 0 & \sin\varphi_2 & \cos\varphi_2 \end{pmatrix} \boldsymbol{r}_2'''$$

3.2. Forward kinematics: 4R arm

The arm consists of a base that rotates about the vertical z-axis by the angle φ_0, followed by three links of lengths R_1, R_2 and R_3 with joint angles φ_1, φ_2 and φ_3 acting in a vertical plane. The first joint sits at height h above the ground, and D_0, D_1 and D_2 are fixed lateral offsets between the joints. All joint angles are signed, with counterclockwise positive; angles drawn in the sketches are unsigned magnitudes. Positions are written in homogeneous coordinates $(1, x, y, z)^\mathsf{T}$.

Figure 3.1 shows the assumed general structure, with $\varphi_0 = 0$ in the top view.

Again we work our way slowly with basis transformations, from the trivial description of the position $\boldsymbol{r}'$ of the end effector to the description $\boldsymbol{r}$ in the general basis:

$$\begin{aligned}\boldsymbol{r} = &\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi_0 & -\sin\varphi_0 & 0 \\ 0 & \sin\varphi_0 & \cos\varphi_0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi_1 & 0 & -\sin\varphi_1 \\ D_0 & 0 & 1 & 0 \\ h & \sin\varphi_1 & 0 & \cos\varphi_1 \end{pmatrix} \\ &\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi_2 & 0 & -\sin\varphi_2 \\ D_1 & 0 & 1 & 0 \\ R_1 & \sin\varphi_2 & 0 & \cos\varphi_2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi_3 & 0 & -\sin\varphi_3 \\ D_2 & 0 & 1 & 0 \\ R_2 & \sin\varphi_3 & 0 & \cos\varphi_3 \end{pmatrix} \boldsymbol{r}'\end{aligned}$$

with

$$\boldsymbol{r}' = \begin{pmatrix} 1 \\ 0 \\ 0 \\ R_3 \end{pmatrix}.$$

4. Inverse Kinematics

4.1. Inverse kinematics: 2R

"Here the second configuration does not exist"

As shown in fig. 4.1a, α, β and γ are the interior angles at the base, the elbow and the target of the triangle formed by the two links and the line from the base to the target, and θ is the angle between the ground and that line.

α, β and γ can be determined via the laws of sine and cosine. Of course, we only need α and β:

$$\alpha = \arctan\left(\frac{\sin(\alpha)}{\cos(\alpha)}\right),$$
$$\beta = \arctan\left(\frac{\sin(\beta)}{\cos(\beta)}\right),$$

where we find $\sin(\alpha)$ and $\cos(\alpha)$ via lemma 2.1. θ we can find directly via

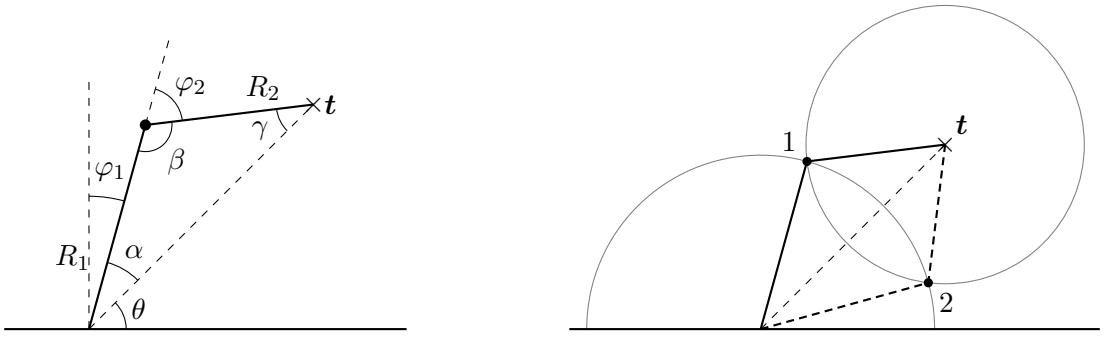
$$\theta = \arctan\left(\frac{t_2}{t_1}\right)$$

with the target $\boldsymbol{t} = (t_1, t_2)^\mathsf{T}$.

Now we have all the ingredients we need to find φ_1 and φ_2 for both configurations:

1. $\varphi_1 = \theta + \alpha - 90°$ and $\varphi_2 = -180° + \beta$,
2. $\varphi_1 = \theta - \alpha - 90°$ and $\varphi_2 = 180° - \beta$.

Hint. Most libraries offer an `atan2`, which resolves all four quadrants.



(a) Configuration 1 with all angles.

(b) Both configurations reaching $\boldsymbol{t}$.

Figure 4.1.: Inverse kinematics of the planar 2R arm. The possible elbow positions are the intersections of the circle of radius R_1 about the base with the circle of radius R_2 about the target.

Reachability. We assume $R_1 > R_2$. Then a target $\boldsymbol{t} = (t_1, t_2)^\mathsf{T}$ is only reachable if

$$R_1 - R_2 \leq |\boldsymbol{t}| \leq R_1 + R_2,$$

where the equals disappears in practice: on the boundary the arm is either fully stretched or fully folded, both configurations coincide, and small errors in the link lengths or rounding in the computation already push the target out of reach.

4.2. Inverse kinematics: 4R arm

The arm and its variables are introduced in section 3.2.

Base angle φ_0. The angle of the x-y-position of the target $\boldsymbol{t} = (1, t_1, t_2, t_3)^\mathsf{T}$ we can compute directly:

$$\varphi_0' = \arctan\left(\frac{t_2}{t_1}\right).$$

But we need to correct this angle by $\delta\varphi_0$, the angle between the arm's x'-axis and the direction to the target (see fig. 3.1). To compute $\delta\varphi_0$ we need the x-y-length of the arm:

$$\delta\varphi_0 = \arcsin\left(\frac{D}{\sqrt{t_1^2 + t_2^2}}\right) \qquad \text{with } D = D_0 + D_1 + D_2.$$

$\delta\varphi_0 \in [0, \frac{\pi}{2})$, so arcsin already gives the unique answer. It also introduces an additional reachability condition:

$$D \leq \sqrt{t_1^2 + t_2^2}.$$

Two configurations for the base angle. As long as the joint angles allow it, one can always create a configuration in which the D_i all point in the positive direction and one in which they all point in the negative direction (fig. 4.2). For a given target this means that two configurations are possible. One of them requires the base angle

$$\boxed{\varphi_0^{(1)} = \varphi_0' - \delta\varphi_0}, \tag{4.1}$$

where φ_0' is the angle of the target and $\delta\varphi_0$ describes the angle of the arm configuration. The other first requires a rotation by π, followed by an inversion of all planar joint angles, $\varphi_i \to -\varphi_i$ for $i = 1, 2, 3$. Afterwards, as fig. 4.2 shows, we have to rotate by $\delta\varphi_0$ further: $\varphi_0^{(2)} = \pi + \varphi_0' + \delta\varphi_0$. Since this number is greater than π,

$$\boxed{\varphi_0^{(2)} = \varphi_0' - \pi + \delta\varphi_0}. \tag{4.2}$$

Reduction to a plane. Once we have φ_0, we can calculate the remaining angles by only considering a plane, specifically the z-x'-plane, where x' is the arm's own horizontal axis shown in fig. 4.2. Why is that the axis? The position in the y'-direction cannot be affected, since the D_i cannot be changed. In the x'-direction, we get changes by moving the links according to the joints' degrees of freedom. The target position in the x'-z-plane is

$$\boldsymbol{t}'^{(1)} = \begin{pmatrix} \sqrt{t_1^2 + t_2^2 - D^2} \\ t_3 \end{pmatrix}.$$

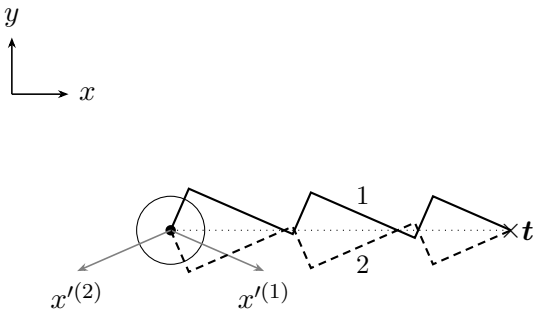
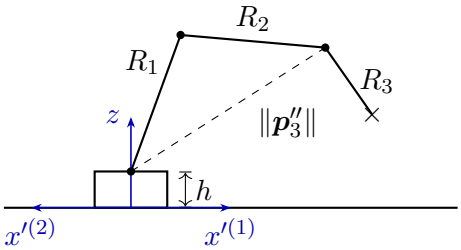
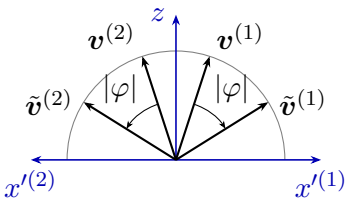


Figure 4.2.: Top view of the two base configurations reaching the same target: configuration 1 (solid) and configuration 2 (dashed), whose x'-axis points away from the links.



(a) The planar problem with the bases of both configurations.



(b) A vector $\boldsymbol{v}$ and its rotation $\tilde{\boldsymbol{v}}$, described in the basis of configuration 1 and in the mirrored basis of configuration 2.

Figure 4.3.: The arm plane of both configurations.

In the case of the second configuration, the x'-axis points in the opposite direction of the links. Therefore we have to invert the sign of the x'-component, where $\varphi_0^{(2)}$ is not explicitly included in the formula:

$$\boldsymbol{t}'^{(2)} = \begin{pmatrix} -\sqrt{t_1^2 + t_2^2 - D^2} \\ t_3 \end{pmatrix}.$$

Wrist position in the x'-z-plane. The 3R problem in the $x'^{(i)}$-z-plane looks identical for both configurations, with the only difference being that the $x'^{(2)}$-axis is mirrored on the z-axis (fig. 4.3a).

Obviously, in both bases the magnitude of the angles does not change; however, the "handedness" of the rotation changes, as fig. 4.3b visualizes. Therefore we only need to compute φ_1, φ_2 and φ_3 for one case and find the angles for the other case by inverting the handedness:

$$\varphi_i^{(2)} = -\varphi_i^{(1)}, \qquad i = 1, 2, 3. \tag{4.3}$$

The target comes with an additional constraint, the angle of vertical approach θ, where $\theta = 0$ implies approaching from straight above (fig. 4.4). With the angle of approach, one can find the position $\boldsymbol{p}_3'$ of the third joint:

$$\boldsymbol{p}_3'^{(1)} = \boldsymbol{t}'^{(1)} + R_3 \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$

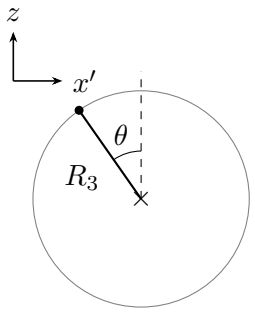


Figure 4.4.: Angle of approach θ.

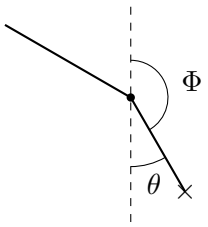


Figure 4.5.: Absolute angle Φ of the last link, measured from the upward vertical, and the angle of approach θ. Both are drawn unsigned; their signs follow from the handedness of the respective basis.

Angles φ_1 and φ_2. Now we can calculate the angles φ_1 and φ_2 the same way as for the 2R arm (section 4.1), with $\boldsymbol{p}_3''$ replacing $\boldsymbol{t}$ (fig. 4.3a), where

$$\boldsymbol{p}_3'' = \boldsymbol{p}_3' - \begin{pmatrix} 0 \\ h \end{pmatrix}$$

is the wrist position relative to the shoulder.

Angle φ_3. With $\Phi = \varphi_1 + \varphi_2 + \varphi_3$ the absolute angle of the last link (fig. 4.5; the angles in the drawing are unsigned), we find for the first configuration

$$\Phi - \theta = \pi, \qquad \varphi_3^{(1)} = \pi + \theta - \varphi_1^{(1)} - \varphi_2^{(1)}.$$

For $\varphi_3 > \pi$, interpret $\varphi_3 \to \varphi_3 - 2\pi$. For the second configuration, (4.3) gives

$$\varphi_3^{(2)} = -\pi - \theta - \varphi_1^{(2)} - \varphi_2^{(2)},$$

which accordingly has to be wrapped into $(-\pi, \pi]$ by adding 2π for $\varphi_3 \leq -\pi$.

Combining the two base configurations with the two solutions of the planar problem (elbow up and elbow down), the 4R arm, in the version used here, can choose between up to four configurations for a given set of constraints, as far as the joint limits allow it.

4.2.1. Extension: 5R – LeRobot style

LeRobot style is basically the 4R arm that was just introduced, but additionally the end effector, a gripper, can rotate about the last link's axis. The end effector of LeRobot is π-symmetric.

The local basis $\boldsymbol{g}_x$, $\boldsymbol{g}_y$, $\boldsymbol{g}_z$ at the gripper is described by $\boldsymbol{g}_z$ aligning with the last link's axis, $\boldsymbol{g}_y$ being the tangent to the base circle at angle φ_0, and $\boldsymbol{g}_x$ being perpendicular to both, with a sign

such that for $\theta = 0$, $\boldsymbol{g}_x$ points in radial direction outwards:

$$\boldsymbol{g}_x = \begin{pmatrix} \cos\theta\cos\varphi_0 \\ \cos\theta\sin\varphi_0 \\ \sin\theta \end{pmatrix}, \qquad \boldsymbol{g}_y = \begin{pmatrix} -\sin\varphi_0 \\ \cos\varphi_0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{g}_z = \begin{pmatrix} -\sin\theta\cos\varphi_0 \\ -\sin\theta\sin\varphi_0 \\ \cos\theta \end{pmatrix}.$$

For the second configuration, the same formulas if one ignores the initial rotation by π, i.e. with $\varphi_0 = \varphi_0^{(2)} - \pi$ and θ unchanged. This however does also neglect that we turn the arm upside down after the initial rotation by π, so the angle χ computed below has to be turned back: $\chi^{(2)} = \chi - \pi$. For the π-symmetric gripper of LeRobot, this correction can be omitted.

Now we can find the line along which the grabbing motion is supposed to take place in the new basis. With $\boldsymbol{l}$ the direction of the grabbing motion, we look for a and b with

$$a\,\boldsymbol{g}_x + b\,\boldsymbol{g}_y = \boldsymbol{l}.$$

This system is overdetermined, and the input is likely noisy; therefore solving it exactly is most likely not possible. Since $\boldsymbol{g}_x$ and $\boldsymbol{g}_y$ are orthonormal, we use

$$a = \boldsymbol{g}_x^\mathsf{T}\boldsymbol{l}, \qquad b = \boldsymbol{g}_y^\mathsf{T}\boldsymbol{l},$$

which is the least-squares solution, i.e. the exact solution for the projection of $\boldsymbol{l}$ onto the $\boldsymbol{g}_x$-$\boldsymbol{g}_y$-plane. Now we can calculate the angle in the $\boldsymbol{g}_x$-$\boldsymbol{g}_y$-plane:

$$\chi = \arctan\left(\frac{b}{a}\right).$$

This is the angle by which we have to rotate the end effector about the last link's axis.

Example: grabbing from above along the x-axis. With $\theta = 0$ and $\boldsymbol{l} = (1, 0, 0)^\mathsf{T}$:

$$\boldsymbol{g}_x = \begin{pmatrix} \cos\varphi_0 \\ \sin\varphi_0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{g}_y = \begin{pmatrix} -\sin\varphi_0 \\ \cos\varphi_0 \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \cos\varphi_0 \\ -\sin\varphi_0 \end{pmatrix},$$

$$\chi = \arctan\left(\frac{-\sin\varphi_0}{\cos\varphi_0}\right) = -\varphi_0.$$

5. Trajectory Planning

5.1. Joint-space vs. Cartesian-space trajectories

5.2. Interpolation

5.3. Velocity and acceleration limits

6. Firmware Implementation

6.1. Software architecture

6.2. Kinematics implementation in C++

6.3. Motor control loop

6.4. Timing and performance

7. Experiments and Validation

7.1. Test setup

7.2. Positioning accuracy and repeatability

7.3. Discussion

8. Mechanical and Electrical Design

8.1. Arm geometry

8.2. Actuators

8.3. Microcontroller and electronics

9. Conclusion and Outlook

9.1. Summary

9.2. Lessons learned

9.3. Outlook

Bibliography

[1] Tilo Arens et al. *Mathematik*. 3rd ed. Berlin, Heidelberg: Springer Spektrum, 2015. DOI: 10.1007/978-3-642-44919-2.

[2] *Curvature*. Wikipedia. URL: https://en.wikipedia.org/wiki/Curvature (visited on 10/08/2026).

[3] Dietmar Gross et al. *Technische Mechanik 2. Elastostatik*. 15th ed. Berlin, Heidelberg: Springer Vieweg, 2024. DOI: 10.1007/978-3-662-68423-8.

A. Source Code

B. Bill of Materials

Part	Type	Source	Qty